

Power Dependence of MEOP Fit Parameters

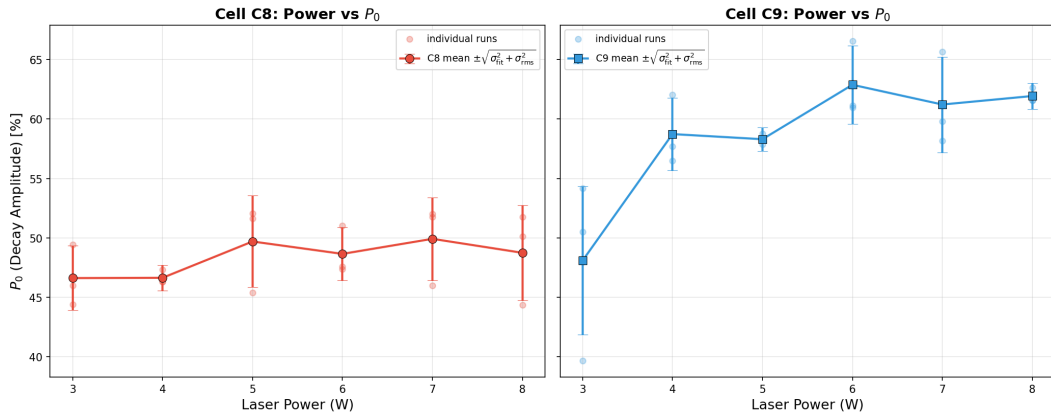
Cells C8 & C9 — 3 W to 8 W

MEOP Analysis

February 26, 2026

Decay Amplitude P_0 vs Laser Power

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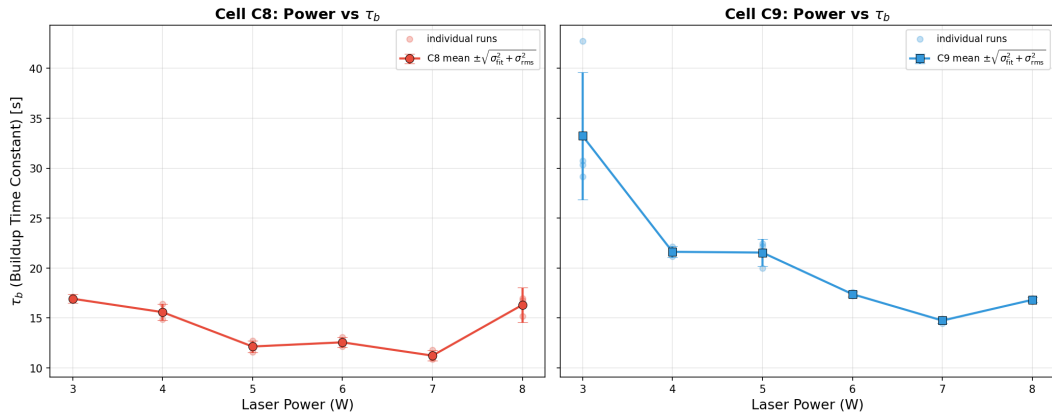


Error bars: $\sigma_{\text{total}} = \sqrt{\sigma_{\text{fit}}^2 + \sigma_{\text{rms}}^2}$

Faded points = individual datagrams

Buildup Time Constant τ_b vs Laser Power

Buildup Time Constant τ_b vs Laser Power



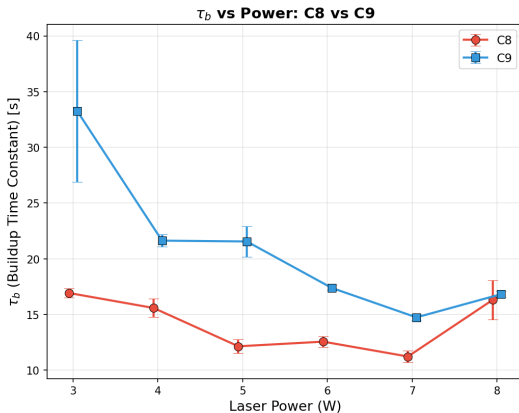
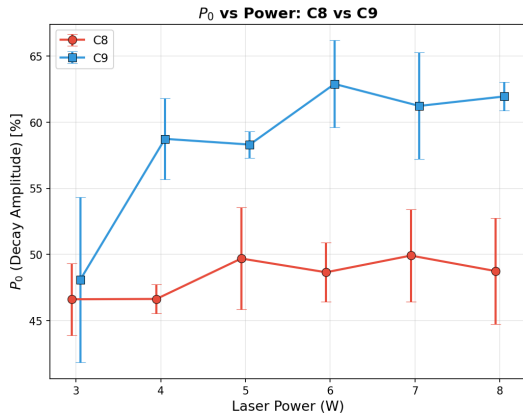
Error bars: $\sigma_{\text{total}} = \sqrt{\sigma_{\text{fit}}^2 + \sigma_{\text{rms}}^2}$ (fit error @ measurement spread)

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C8 vs C9 Comparison

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Error bars: $\sigma_{\text{total}} = \sqrt{\sigma_{\text{fit}}^2 + \sigma_{\text{rms}}^2}$ (fit error $\oplus$ measurement spread)

● C8 ● C9

Error: $\sigma_{\text{fit}} \oplus \sigma_{\text{rms}}$ added in quadrature

Summary of Observations

Decay Amplitude P_0

- **C8**: Relatively flat ($\sim 47\text{--}50\%$) across all powers
- **C9**: Increases from $\sim 48\%$ (3 W) to $\sim 63\%$ (6 W), then plateaus
- C9 consistently higher than C8 above 4 W

Buildup Time τ_b

- Both cells: τ_b decreases with increasing power (faster buildup)
- **C9** starts much higher ($\sim 33\text{ s}$ at 3 W) vs **C8** ($\sim 17\text{ s}$)
- Convergence at high power ($\sim 16\text{ s}$ at 8 W)

All error bars: $\sigma_{\text{total}} = \sqrt{\sigma_{\text{fit}}^2 + \sigma_{\text{rms}}^2}$ (fit uncertainty $\oplus$ run-to-run spread, independent errors)